

CHAPTER 2

Activities & Lab Packet

Energy

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 2. Works with both the Standard and Easy Read versions of the textbook.



Georgia Standard S8P2

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1 Pendulum Energy Swap

LAB

OBJECTIVE

See KE and PE trade places in a swinging pendulum.

TIME

25–35 minutes

STANDARDS

S8P2.a, S8P2.b

Materials (per group of 2–3)

- 1 piece of string (about 50 cm long)
- 1 small heavy object (washer, nut, or fishing weight)
- 1 ring stand or sturdy edge to tie the string to (a doorknob or table edge works)
- 1 ruler or meter stick
- 1 stopwatch (phone timer is fine)
- Student data sheet (next page)

Setup

Tie the weight to one end of the string. Tie the other end to your ring stand or edge so the weight hangs freely. This is your pendulum. Make sure it can swing without hitting anything.

Student Instructions

1. Pull the pendulum back to a small starting height (about 10 cm above its lowest point). Hold it steady.
2. At this point, the pendulum is NOT moving. Where is the energy? Record your answer.
3. Let go. Watch the pendulum swing back and forth.
4. At what point is the pendulum moving the fastest? Where in the swing is this?
5. At what point does the pendulum stop momentarily? Where in the swing is this?
6. Time how long it takes the pendulum to swing 10 times. Record this.
7. Now try it again with the pendulum pulled back to 20 cm (higher starting point). What changes?
8. Now try the same height (10 cm) but with a heavier weight. What changes?

TEACHER NOTES

The key concept: At the top of each swing, the pendulum momentarily stops — all PE, no KE. At the bottom of the swing, it's moving fastest — all KE, minimum PE. Energy is being traded back and forth.

Discussion at end: "Why does the pendulum eventually slow down and stop? Where did the energy go?" (Answer: friction and air resistance turn the kinetic energy into heat and sound.)

Extension: Try the heavier weight at the same height. Students often expect it to swing faster — it doesn't! Period depends on string length, not mass.

Pendulum Energy Swap — Data Sheet

Name: _____

Period: _____

Part A: Identify the energy

At the highest point (just before letting go), the pendulum has mostly: ☐ KE ☐ PE

At the lowest point of its swing, the pendulum has mostly: ☐ KE ☐ PE

At the highest point on the OTHER side of the swing, it has mostly: ☐ KE ☐ PE

Part B: Data Table

Trial	Starting Height	Mass	Time for 10 swings
1 (original)	10 cm	1 weight	
2 (higher)	20 cm	1 weight	
3 (heavier)	10 cm	2 weights	

Why does the pendulum slow down and eventually stop?

Did changing the mass change the time? Explain.

2 Marble Coaster Engineering

LAB

OBJECTIVE

Design a roller coaster track that uses energy transformation.

TIME

45–60 minutes (can be split across 2 days)

STANDARDS

S8P2.a, S8P2.b, S8P2.c

Materials (per group of 3–4)

- 2–3 lengths of foam pipe insulation, cut in half lengthwise (makes a great track) — or pool noodles cut in half
- 5–10 marbles or steel balls
- Masking tape
- Cardboard boxes, books, or chairs (for height supports)
- A plastic cup (the "end zone" target)
- Ruler or measuring tape
- Student design sheet

The Challenge

Each group builds a marble track that:

- Starts at a measurable height
- Has at least **2 hills** the marble has to roll over
- Has at least **1 loop or curve**
- Ends with the marble dropping into the cup at the end

Student Instructions

1. Sketch your design before building. Label the "PE high point" and the "KE high point" of your track.
2. Build your track. Tape the foam pieces together. Use boxes for supports.
3. Test it! If the marble doesn't make it to the cup, figure out why and adjust.
4. Once it works, measure: starting height, ending height, and total length.
5. Find the highest point AFTER the first drop. Why does this second hill have to be LOWER than the first?
6. Try a heavier marble. Does it still work?

TEACHER NOTES

The key engineering concept: Each hill after the first MUST be lower than the previous one. Why? Because some PE is converted to heat/sound at every turn, leaving less energy for the next hill.

Failure is learning: Don't intervene too early when a track fails. Let groups troubleshoot — that's where the real physics learning happens.

If you can't get pipe insulation: Cardboard tube halves work. So do cardboard strips bent into a U-shape and taped together.

Optional competition: Award "most creative loop," "longest track," "smoothest run," etc. Avoid only ranking by speed.

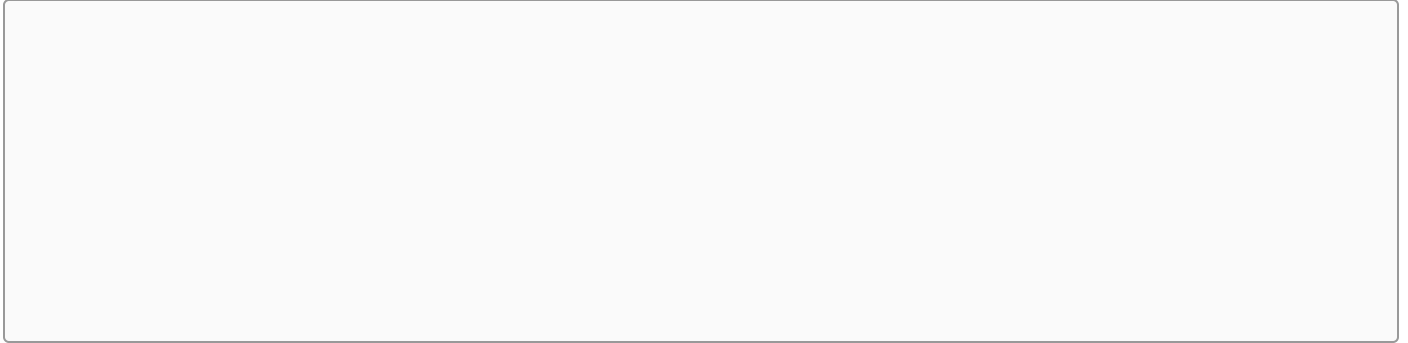
Marble Coaster — Engineering Design Sheet

Name: _____

Period: _____

Step 1: Sketch your design

Draw your planned track. Label the start, end, and where PE and KE are highest.



Step 2: Record measurements (after building)

Starting height of marble:		cm
Height of the SECOND hill:		cm
Total length of track:		cm
Did the marble make it to the cup?	☐ YES ☐ NO	

Why does the second hill HAVE to be lower than the starting hill?

3 Heat Transfer Race

LAB

OBJECTIVE

Compare how heat moves through different materials.

TIME

30 minutes

STANDARDS

S8P2.d

Materials (per group)

- 1 cup of HOT tap water (very warm, but not dangerous)
- 1 metal spoon
- 1 plastic spoon
- 1 wooden craft stick (popsicle stick)
- 3 small pats of butter (or 3 mini marshmallows, or 3 mini chocolate chips)
- Stopwatch (phone timer)

Setup

Each group gets the cup of hot water. The three utensils stand up inside the cup. Near the TOP of each utensil (outside the water), stick a small piece of butter (or marshmallow). The butter will tell us when heat reaches the top.

Student Instructions

1. Predict: which utensil will let heat travel up the fastest? Rank them 1, 2, 3 (1 = fastest).
2. Put a small piece of butter near the top of each utensil. They should be equal distance from the water line.
3. Start the timer when all three utensils are in the hot water.
4. Watch carefully. Time how long it takes for the butter on EACH utensil to start melting/dripping.
5. Record your results. Which one happened first?

TEACHER NOTES

Expected result: Metal melts butter fastest (best conductor). Plastic is slowest or doesn't melt at all (insulator). Wood is in the middle but still slow.

Safety: Use HOT tap water, not boiling. Students should not pour their own water — pre-fill cups.

The big concept: All three are showing CONDUCTION (heat moving through direct contact). The difference is how well each material lets heat flow through. This is also a great segue into "why are pot handles wood/plastic instead of metal?"

Discussion bonus: If you placed your hand 5 cm ABOVE the hot water without touching it, what type of heat transfer would you feel? (Convection from rising warm air, plus a little radiation.)

Heat Transfer Race — Data Sheet

Name: _____

Period: _____

My PREDICTION (1 = fastest, 3 = slowest):

Metal spoon: _____ Plastic spoon: _____ Wooden stick: _____

RESULTS:

Utensil	Time for butter to melt	Material type (conductor or insulator?)
Metal spoon		
Plastic spoon		
Wooden stick		

What type of heat transfer happened in this experiment?

☐ Conduction ☐ Convection ☐ Radiation

Why are pot handles usually made of plastic or wood instead of metal?

4 KE & PE Practice

WORKSHEET

OBJECTIVE

Apply the relationships between KE-mass-speed and PE-mass-height.

TIME

20 minutes

STANDARDS

S8P2.a

TEACHER NOTES

This worksheet focuses on the *relationships*, not the formulas. Students don't need to plug numbers into $KE = \frac{1}{2}mv^2$ — they just need to compare which object has more KE or PE based on mass, speed, and height. This matches what S8P2.a expects.

KE & PE — Who Has More?

Decide which object has MORE energy. Circle your answer.

Name: _____

Period: _____

Part A: Kinetic Energy

- Both moving at 30 mph: a bicycle OR a school bus? → **Bicycle / Bus**
- Both have mass of 5 kg: one moving 2 m/s, one moving 10 m/s? → **2 m/s / 10 m/s**
- A car driving 60 mph north OR the same car driving 60 mph south? → **North / South / Same**
- A bowling ball rolling at 5 m/s OR a soccer ball rolling at 5 m/s? → **Bowling / Soccer**

Part B: Potential Energy

- Both 2 kg: one on a 5 m shelf, one on a 10 m shelf? → **5 m / 10 m**
- Both at 3 m high: a 1 kg book OR a 10 kg book? → **1 kg / 10 kg**
- A 5 kg cat on the floor OR a 5 kg cat on the table? → **Floor / Table**
- A roller coaster car at the top of the hill OR at the bottom of the hill? → **Top / Bottom**

Part C: KE and PE Together

- At what point on a roller coaster is PE highest? _____
- At what point on a roller coaster is KE highest? _____
- What happens to the PE as a ball falls from a shelf? _____
- What happens to the KE as a ball falls from a shelf? _____

Part D: Big Idea

If a pendulum starts with 100 joules of PE at the top of its swing, how many joules of KE will it have at the very bottom of its swing? _____

(Hint: think about conservation of energy!)

ANSWER KEY

1) Bus (more mass) | 2) 10 m/s (more speed) | 3) Same (KE doesn't depend on direction) | 4) Bowling (more mass) | 5) 10 m (more height) | 6) 10 kg (more mass) | 7) Table (more height) | 8) Top (more height) | 9) At the top of the highest hill | 10) At the bottom of the biggest drop | 11) PE decreases | 12) KE increases | **Part D:** 100 J (energy is conserved — PE becomes KE)

5 Energy Transformation ID

WORKSHEET

OBJECTIVE

Identify the form energy changes from and to in everyday objects.

TIME

15–20 minutes

STANDARDS

S8P2.c

Energy Transformation ID

Write the starting energy → ending energy for each. Use this word bank.

Name: _____

Period: _____

WORD BANK: chemical, electrical, light, sound, thermal (heat), kinetic (motion), potential (stored), nuclear

#	Object or Event	Starting Energy	Ending Energy
1	A lightbulb turning on		
2	Striking a match		
3	A microwave heating soup		
4	A speaker playing music		
5	A solar panel		
6	A wind-up toy moving		
7	You eating a sandwich, then walking		
8	A campfire		
9	A nuclear power plant		
10	Plants growing in the sun		

Bonus Challenge: Find one object that does **THREE** transformations in a row. Write it out.

ANSWER KEY

1) Electrical → Light + Heat | 2) Chemical → Light + Heat | 3) Electrical → Light (microwaves) → Heat | 4) Electrical → Sound | 5) Light → Electrical | 6) Potential (elastic) → Kinetic | 7) Chemical → Kinetic + Heat | 8) Chemical → Light + Heat | 9) Nuclear → Heat → Kinetic → Electrical | 10) Light → Chemical

Bonus: A flashlight = Chemical (battery) → Electrical → Light. A car = Chemical (gas) → Heat → Kinetic. Accept any reasonable 3-step chain.

6 Conduction, Convection, or Radiation?

WORKSHEET

OBJECTIVE

Classify everyday examples of heat transfer.

TIME

15 minutes

STANDARDS

S8P2.d

Conduction, Convection, or Radiation?

Decide which type of heat transfer is happening. Circle your answer.

Name: _____

Period: _____

#	Situation	Heat Transfer Type
1	You touch a hot pan handle and burn your hand	Cond. / Conv. / Rad.
2	You feel the Sun warm on your face on a cold day	Cond. / Conv. / Rad.
3	Hot air rises from a heating vent and warms the room	Cond. / Conv. / Rad.
4	A spoon left in a bowl of hot soup gets hot	Cond. / Conv. / Rad.
5	Boiling water in a pot bubbles and circulates	Cond. / Conv. / Rad.
6	You warm your hands by holding them NEAR (not touching) a campfire	Cond. / Conv. / Rad.
7	An ice cube melts in your hand	Cond. / Conv. / Rad.
8	Warm ocean water rises and cooler water sinks	Cond. / Conv. / Rad.
9	A toaster turns red and heats your bread without touching it	Cond. / Conv. / Rad.
10	A hot air balloon rises into the sky	Cond. / Conv. / Rad.
11	Heat from the Earth's core travels through molten rock	Cond. / Conv. / Rad.
12	A microwave heats up frozen leftovers	Cond. / Conv. / Rad.

Short Answer: Which type of heat transfer can travel through empty space (a vacuum)? Why?

ANSWER KEY

1) Conduction | 2) Radiation | 3) Convection | 4) Conduction | 5) Convection | 6) Radiation | 7) Conduction | 8) Convection | 9) Radiation | 10) Convection | 11) Convection (molten rock is a fluid!) | 12) Radiation (microwaves are EM waves)

Short Answer: Radiation. It travels as electromagnetic waves and doesn't need a medium. That's how the Sun's energy reaches Earth through space.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain energy concepts in writing.

TIME

15–30 minutes each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Bottled Sunshine.** The textbook says your lunch is "bottled sunshine." Explain in your own words what this means. Trace the energy from the Sun all the way to your muscles.
2. **My Favorite Energy.** Pick one form of energy that interests you. Describe it and give two real-life examples of where you see it.
3. **Roller Coaster Day.** You're on a roller coaster. Describe how your energy changes from the start of the ride to the end. Use the words "kinetic" and "potential."

Level 2 — Intermediate (3 paragraphs)

4. **The Pendulum Story.** Pretend you ARE a pendulum. Tell the story of one swing — from being pulled back, to swinging down, to going up the other side. Describe what's happening to your KE and PE the whole way through.
5. **Where Did the Energy Go?** A bouncing ball eventually stops bouncing. The Law of Conservation of Energy says energy is never destroyed. So where did the ball's energy go? Explain using friction, heat, and sound.
6. **Cooking with All Three.** Pick a meal you love that's cooked on a stove. Explain how conduction, convection, AND radiation all play a role in making that meal.

Level 3 — Advanced (essay)

7. **The Power Plant.** Most electricity comes from power plants that burn fuel. Trace the energy transformations from buried coal all the way to your phone charger. Then argue: at what point is energy "wasted"? Where does that wasted energy actually go?
8. **If Energy Could Be Destroyed.** Imagine the Law of Conservation of Energy was not true and energy COULD disappear. Write about three ways your world would be different. Use specific examples.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific thinking through verbal reasoning.

USE

Warm-ups, exit tickets, Socratic seminar

STANDARDS

Cross-cutting

TEACHER NOTES

These are designed for "think-pair-share" format. Some are gotcha-style and intentionally counterintuitive — those work great for building scientific argumentation skills. Push students to defend their answers with evidence from the chapter.

Quick Warm-Ups (1–2 min each)

1. Does a sleeping cat have any energy? Why or why not?
2. Can you have temperature without thermal energy?
3. You rub your hands together — where does the heat come from?
4. If a roller coaster started with 100 J of PE, can it end with 110 J of KE? Why or why not?
5. How is the Sun's heat able to reach you when most of the path between you and the Sun is empty space?

Deeper Discussions (5–15 min each)

6. **The Pencil Drop Paradox.** If energy can never be destroyed, where does the kinetic energy of a pencil go when it falls and hits the floor? Why doesn't it bounce back up to its starting height?
7. **Hot Tub vs. Boiling Cup.** A bathtub of warm water has MORE thermal energy than a small cup of boiling water — even though the cup is way hotter. How is this possible?
8. **Why Don't Pendulums Swing Forever?** A pendulum in a perfect vacuum with no friction would swing forever. Why doesn't this happen in real life?
9. **Cold Doesn't Exist.** Some scientists argue that "cold" isn't a real thing — only the ABSENCE of heat. Do you agree? Defend your answer using what you know about thermal energy.
10. **The Best Way to Cool a Room.** Should you put a fan in the window blowing IN, or blowing OUT? Both are correct in different situations. Explain when each would work best using convection.

9 Build an Energy Transformer

GROUP PROJECT

OBJECTIVE

Design and build a chain of at least 4 energy transformations.

TIME

2–3 class periods

STANDARDS

S8P2.b, S8P2.c

Project Description

This is like a mini Rube Goldberg machine. Each group designs a chain reaction that uses at least **4 energy transformations** to complete a simple task — like popping a balloon, ringing a bell, or knocking down a cup.

Requirements

- At least **4 distinct energy transformations**
- Must include at least one of: KE $\leftrightarrow$ PE swap, chemical energy use, heat transfer, sound, or light
- The chain must complete its task automatically once started
- The group must label EACH transformation on their final diagram

Easy-to-find Materials Ideas

Materials	Energy Type Available
Marbles, dominoes, toy cars on ramps	KE $\leftrightarrow$ PE
Mousetraps, rubber bands, springs	Elastic PE $\rightarrow$ KE
Flashlight, small lamp	Electrical $\rightarrow$ Light
Wind-up toys, pendulums	Stored $\rightarrow$ Motion
Magnets, paperclips, iron objects	Magnetic force $\rightarrow$ motion
Bells, whistles, music makers	Motion $\rightarrow$ Sound
Hot water + thermometer	Thermal energy demos

Required Deliverables

1. **Build the machine** and demonstrate it working.
2. **Create a labeled diagram** showing each transformation in the chain.
3. **Present to the class.** Each group member explains one of the transformations.
4. **Final write-up:** Where do you think energy is being "lost" to heat or sound? Where is it going?

TEACHER NOTES

Differentiation: Allow lower-readiness groups to use just 3 transformations. Push advanced groups to do 6+.

Suggested rubric: Number of transformations (30%), labels accuracy (25%), creativity (15%), presentation (15%), write-up (15%).

Time savers: Have a "parts bin" of marbles, dominoes, string, paperclips, rubber bands, paper cups, and tape ready for kids to use.

Backup plan: If physical materials aren't available, students can DRAW their machine instead. The transformations are still the main learning outcome.

10 Energy Vocabulary Memory Match

[REVIEW GAME](#)

OBJECTIVE

Review chapter vocabulary in a low-stress game format.

TIME

20–30 minutes

STANDARDS

S8P2 cross-cutting

Setup

Print and cut out the cards on the next page (one set per group of 2–3). Each set should be face-down in a grid. Students take turns flipping 2 cards at a time, trying to match a term with its definition. If they match, they keep the pair. Most pairs at the end wins.

Card Set — Print & Cut (20 cards = 10 pairs)

ENERGY	The ability to do work or cause change
KINETIC ENERGY	The energy of motion
POTENTIAL ENERGY	Stored energy based on position
JOULE	The unit used to measure energy
CONDUCTION	Heat transfer through direct contact
CONVECTION	Heat transfer through fluid currents
RADIATION	Heat transfer through EM waves (no medium needed)
TEMPERATURE	Average kinetic energy of the particles
THERMAL ENERGY	The TOTAL energy of all moving particles
CONSERVATION OF ENERGY	Energy cannot be made or destroyed, only changed

TEACHER NOTES

Print prep: Print on heavier paper or cardstock if possible. Laminate if you want a reusable set.

Variations:

- **Speed Match:** All cards face up — first to find and match all pairs wins.
- **Charades version:** Pull a vocab card and act it out for your group to guess.
- **Add more cards:** Include "chemical energy," "mechanical energy," "elastic PE," "law of conservation of mass" for a 30-card set.

Easy adaptation: For struggling readers, leave the chapter vocabulary list nearby so they can reference it during the game.